Mariem Kbaier

Final-Year Computer Engineering Student | AI & Data | Tunis, Tunisia

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Professional Summary

Seeking an end-of-studies internship (PFE) in AI & Data. As a final-year computer engineering student at ENIS, I am interested in designing intelligent solutions, analyzing data and solving real-world problems with new technologies.

Experience

- **AgriMent** – Multi-agent AI platform for agricultural decision support repo
  Talan Tunisia – Summer Camp *Tunis | July – August 2026*
  LangGraph, Python, FastAPI, Mistral, Hybrid RAG (BM25 + RRF), Qdrant, ChromaDB, PyTorch, TempCNN, Sentinel-2, PostgreSQL/PostGIS, React, Docker
  - Designed a 9-microservice decision-support platform for French farmers and buyers (regulation, market, crop monitoring) with **dynamic multi-agent routing**, a supervisor and agent-initiated handoffs. **2nd out of 10 teams**.
  - Built a regulatory **hybrid RAG** pipeline (Mistral embeddings + BM25 + RRF) over Légifrance, FranceAgriMer and CERFA forms; the best configuration reaches **0.626 faithfulness** (+14 % vs. no RAG), with RAGAS scores validated by expert evaluations.
  - Fine-tuned a **TempCNN** crop classifier on Sentinel-2 imagery (**80.3 % accuracy**) and designed a graded-confidence architecture (LLM-grounded / ML-predicted / deterministic) with a numeric audit guardrail, aligned with the EU AI Act transparency rules.
- **Full-stack technical internship**
  BNS Engineering – Developer Intern *Tunis | June – August 2025*
  Spring Boot, REST APIs, React.js, MySQL, Docker, Git, Agile/Scrum
  - Developed a secured **Spring Boot REST API** in a microservices architecture, with a React.js frontend, MySQL and Docker deployment, for a self-service **service catalog** (Agile/Scrum team).
  - Integrated a **semantic chatbot** for natural-language search in the catalog and user assistance.

Projects

- **AI Agent for Company Performance Analysis** – KPI & Power BI *(Academic project, ENIS)* repo
  Python, Pandas, LLM, Power BI, DAX
  - Designed an **AI agent** that analyzes company KPIs, **detects notable changes**, identifies the main product or region involved and writes a **concise conclusion** in natural language.
  - Separated **deterministic computation** (Python/Pandas) from LLM writing to guarantee reliable figures; results displayed and **validated in a Power BI dashboard**.
- **Insurance Data Analytics** – Data engineering on Microsoft Fabric repo
  Microsoft Fabric, Data Pipeline, OneLake, Lakehouse, PySpark, SQL, Power BI
  - End-to-end **ETL/ELT pipeline**: ingestion via Data Pipeline, OneLake/Lakehouse storage, **PySpark** transformation and a **Power BI dashboard** of insurance indicators.
- **VORTEX** – Multi-agent BDD testing *(submission in progress, ICAART 2027)* repo
  LangGraph, LangChain, Groq API, ChromaDB, Java, Spring Boot, Cucumber, RestAssured, JaCoCo, Maven
  - Built a framework that turns business requirements into **executable, self-healing test suites** for Spring Boot microservices.
  - Designed a **multi-agent LangGraph pipeline** combining RAG over BDD/JUnit scenarios and a self-repair loop: scenario coverage **5 %→100 %**, pass rate **32 %→81.2 %**.
  - Benchmarked several LLMs on output quality and token cost to select the best model for each agent.
- **End-to-End AI-Driven QA Pipeline** *(Academic project, ENIS, pair project)*
  AI orchestration, test generation, CI/CD, test management
  - Designed an AI-orchestrated end-to-end QA chain: requirements → test cases → export to a test management tool → automated tests → **continuous integration** run → summary report, with a focus on **reproducibility**.
- **Job Matcher** – Multi-agent AI CV/job matching system *(Personal project – Talan Tunisia)* repo
  LangGraph, MCP (FastMCP), Apify API, Groq (Llama 3.3 70B), Sentence-Transformers, ChromaDB, Streamlit
  - Developed a candidate–job matching pipeline with evidence-backed explanations, fed with real job postings through the **Apify API** exposed as an **MCP tool**.

  - Implemented **deterministic scoring** (embedding similarity + skills overlap) independent of LLM generation, with a **code-verified citation system** linking each explanation to the source text.
  - Validated model changes with a **golden dataset** integrated into CI.
- **Company Logo Detection** – Computer vision repo
  Python, YOLOv8, Ultralytics, PyTorch, Google Colab (GPU)
  - Fine-tuned **YOLOv8s** to detect **10 company logos** on an annotated dataset: **mAP50 of 0.995** on the validation set, with fast GPU inference.
- **Customer Behavior Analysis & Churn Prediction** repo
  Python, Pandas, Scikit-Learn, XGBoost, K-Means, DBSCAN, Random Forest, Flask
  - Built an end-to-end ML pipeline on **4,372 customers**: churn prediction, segmentation and spending forecasting.
  - **XGBoost** for churn (**F1: 0.955**), **K-Means/DBSCAN** for segmentation and **Random Forest** for spending.
  - Deployed the models through a **Flask REST API** with an interactive dashboard.
- **FogWatch** – Collaborative detection of distributed SSH attacks *(Academic project, team)* repo
  Suricata, MQTT (Mosquitto), K3s, Docker, Tailscale, nftables, Python, scikit-learn
  - Designed a **collaborative IDS** on a fog architecture (3 sites): each site detects with Suricata and shares its observations over **MQTT**; a correlator spots "low and slow" SSH attacks, invisible from a single site, and blocks the IP on every site.
  - Deployed the agents on **K3s** (shared manifests) and prototyped a **Random Forest** SSH flow classifier trained on CICIDS2017 (F1 $\approx$ 0.96 in cross-validation).

Skills

- **AI & Machine Learning:** LangGraph, LangChain, AI agents, RAG (hybrid search, BM25/RRF), multi-agent systems, MCP, LLM evaluation (RAGAS), Sentence-Transformers, ChromaDB, Qdrant, PyTorch, scikit-learn, XGBoost, computer vision (YOLOv8)
- **Data & BI:** Pandas, SQL, Power BI, DAX, KPI analysis, PostgreSQL/PostGIS, MySQL, NoSQL
- **MLOps & Cloud:** Docker, GitHub Actions (CI/CD), Git, AWS (Lambda, DynamoDB, IoT Core, S3)
- **Languages & frameworks:** Python, Java, JavaScript, SQL; FastAPI, Flask, Spring Boot, React.js, Streamlit
- **Concepts:** microservices, BDD/testing, MLOps, Agile/Scrum

Education

- **ENIS** – Engineering degree in Computer Science — Sfax, Tunisia | 2024 – Present
- **IPEIT** – Preparatory cycle (MPI) — Tunis, Tunisia | 2021 – 2024

Honors & Activities

- **2nd place – Talan Summer Camp 2026:** 2nd out of 10 teams for AgriMent, a generative-AI-based agricultural decision-support platform.
- **1st place – Non-Technical Challenge:** Represented the IEEE PES Chapter – ENIS Student Branch as **Webmaster**.
- **1st place – TEC.Week 3.0:** Tunisian Engineering Congress Week.

Certifications, Involvement & Languages

- **Certifications:** Agentic AI Certified Foundations Associate (Oracle) | LangGraph – Multi-Agent Systems & Building Scalable Agentic Systems (DataCamp) | Cisco C Advanced | Scrum Fundamentals (SCRUMstudy) | Attendance Hashgraph Developer (The Hashgraph Association)
- **Involvement:** IEEE member – Pitch events organizer
- **Languages:** Arabic (native) | English (C1) | French (B2)